

SLIDE 1: TITLE & HOOK

Unify the Data. Remember the Story. Recover with Fairness.

CORE MISSION

Core Philosophy

- ▣ Paradigm Shift:
From aggressive dunning to empathetic, customer-centric financial rehabilitation.
- ▣ Sustainable Recovery:
Optimizing interaction appropriateness rather than coercive pressure.

INNOVATIONS

Dual Innovations

- ▣ Golden Financial ID:
Persistent cross-system identifier resolving fragmented account records.
- ▣ Collection Memory:
Longitudinal interaction event log ensuring agents never start from zero.

CAPABILITY

Frontline Empowerment

- ▣ Agent Assist Cockpit:
Real-time decision intelligence with explainable Next Best Actions.
- ▣ Customer 360:
Complete exposure, delinquency, and income picture on a single pane.

VALIDATION

Dataset Grounding

- ▣ Scale & Scope:
68,162,322 records across 1,000,000 synthetic customers at Maple Bank.
- ▣ Rigorous Validation:
Stress-tested across 35 official benchmark scenarios with 100% coverage.

SLIDE 2: THE CORE PROBLEM

The Siloed Collections Dilemma Across 5 Disparate Systems

SYSTEM SILOS

The Data Disconnect

❑ Fragmented Keys:

CRM ID != Card System Ref != Borrower Loan # != Core Banking CIF.

❑ Blind Operations:

Collections operates on account-level silos with zero cross-product view.

AGENT STRUGGLE

The Frontline Burden

❑ Zero Memory:

Every call starts from scratch; borrowers re-tell their story endlessly.

❑ Call Abandonment:

Frustrated customers hang up; 68% drop rate on repeat dunning calls.

FLAWED LOGIC

The Flawed Question

❑ Traditional Dunning:

"How much does this customer owe on this account right now?"

❑ Missed Reality:

Ignores whether the customer can sustainably afford to pay today.

IMPACT

Business Consequences

❑ Avoidable Charge-Offs:

Accounts default unnecessarily due to rigid payment demands.

❑ Damaged Relationships:

Long-term banking value destroyed by abrasive collection methods.

The Golden Financial Identity Layer

ARCHITECTURE

Conceptual Identity Layer

- ▣ Persistent Key:
Inspired by persistent IDs (like PAN); unifies disparate customer accounts.
- ▣ Bank-Controlled:
Conceptual data layer connecting records without creating a bloated god DB.

RESOLUTION

Entity Resolution Engine

- ▣ Deterministic Pass:
Exact matching on normalized Phone, Email, and DOB (Confidence = 1.0).
- ▣ Probabilistic Pass:
Jaro-Winkler similarity on name/address for fuzzy account linking.

SAFETY GATE

Safety & Review Queue

- ▣ Zero Silent Merges:
Matches below 0.85 confidence routed to human review (10,365 flagged).
- ▣ Lineage Retention:
Every field preserves source_table, original timestamp, and confidence.

PERFORMANCE

Production Performance

- ▣ 1M Customers Unified:
1,020,000 CRM profiles clustered into 1,001,879 unique GID entities.
- ▣ 14.4s Execution:
1,788,432 cross-source records resolved in under 15 seconds.

SLIDE 4: INNOVATION PILLAR 2

Collection Memory & Longitudinal Context

DATA STORE

Longitudinal Event Log

▣ 3.58M Events:

Unified event history from contact_history, promises, notes, transcripts

▣ Chronological Stream:

Time-stamped audit of what happened, when, and what followed.

INTELLIGENCE

Five Golden Questions

▣ Question 1 & 2:

What happened before? What is happening right now?

▣ Question 3, 4 & 5:

What has worked? What has NOT worked? What should I consider next?

NLP ENGINE

NLP Hardship Extraction

▣ Unstructured Signals:

Scanned 760,000+ agent notes and transcripts for distress keywords

▣ 84,289 Cases Uncovered:

Identified hidden hardship under-reported by legacy manual flags (~60%).

SAFEGUARDS

Protective Safeguards

▣ Protection First:

Hardship strictly triggers payment pauses, forbearance, and specialist care

.

▣ Never Adverse:

Explicitly prohibited from being used to penalize or accelerate recovery.

BRANCH A

Branch A Exposure Analysis

▣ Cross-Product Debt:

Aggregates credit card balances, personal loans, mortgages, overdraft.

▣ DTI Metric:

Total debt exposure divided by verified annual income, scaled 0 to 1.

BRANCH B

Branch B Cash-Flow Stress

▣ Cash Inflow:

Verified monthly salary deposits and payroll credit frequency.

▣ Liquidity Shocks:

Tracks payroll delay days, NSF occurrences, and overdraft spikes.

BRANCH C

Branch C: Payment Behaviour

▣ Promise Integrity:

Historical Promise-to-Pay fulfillment rate over rolling 24 months.

▣ Delinquency Velocity:

Rate of progression through delinquency buckets (1-30, 31-60, 90+).

AFFORDABILITY

Affordability Assessment

▣ Debt Service Ratio (DSR):

True monthly obligations divided by monthly cash inflow.

▣ Arrangement Capacity:

Deterministic scoring: YES (sustainable), MARGINAL (step-up), NO.

POLICY RULES

5 Policy Rules Enforced

▣ Rule 1: Hardship Safeguard:

Medical/job-loss signals route directly to human specialist review.

▣ Rule 2: DPD > 90 Escalation:

Late-stage accounts trigger senior supervisor evaluation.

DECISIONING

Affordability Decisioning

▣ Rule 3: Sustainability Gate:

Cannot support arrangement -> Routed to hardship assistance.

▣ Rule 4: Marginal Capacity:

Tight budget -> Tailored, affordable step-up payment plan proposal.

EARLY STAGE

Early Stage Treatment

▣ Rule 5: Digital Reminder:

DPD <= 30 with GOOD/FAIR health receives non-intrusive reminder.

▣ Least Intrusive:

Avoids costly agent calls for minor administrative oversights.

HUMAN CONTROL

Frontline Agent Control

▣ Human-in-the-Loop:

Agent retains final decision: Accept, Modify, or Override recommendation.

▣ 100% Explainable:

Plain-English explanation, cited policy rule, and evidence on every case.

The 4 Governed Layers Aligned to Hackathon Technical Scope

LAYER 1

Layer 1: Data Product Factory

▣ Vectorized Ingestion:

DuckDB SQL engine ingesting 31 tables (68.1M rows) in 32 seconds.

▣ Open Data Contract

ODCS v3.0.0 contract (DC-COLL-360) enforcing schema and SLAs.

LAYER 2

Layer 2: Insight & NLP Platform

▣ NL-to-SQL Querying:

Supervisor queries converted to executable SQL with instant execution.

▣ Refusal Guardrails

Strict refusal of PII exfiltration and protected attribute queries.

LAYER 3

Layer 3: Features & Memory

▣ Feature Store:

Unified feature engineering for exposure, cashflow, and behaviour.

▣ Longitudinal Memory:

Time-aware event store driving Five Golden Questions context.

LAYER 4

Layer 4: AI Decisioning Cockpit

▣ Agent Assist Interface:

Interactive Streamlit Cockpit with 6 integrated panels for cases.

▣ Smart Payment Plans:

3 pre-calculated, affordability-verified arrangement options.

Grounded Validation Across 35 Official Benchmark Scenarios

BENCHMARK

100% Benchmark Coverage

▣ Complete Evaluation:

All 35 benchmark questions (dev + test splits) evaluated and verified.

▣ Automated Submission:

benchmark_answers.csv generated with exact SQL and document citations.

ACCURACY

Ground-Truth Accuracy

▣ BQ-035 Open Cases:

Matches official benchmark to the exact integer across all 7 buckets.

▣ Incident Root Cause:

BQ-030 correctly traces call drop to dialer outage (INC-0923).

GUARDRAILS

Refusal Compliance

▣ Out-of-Scope (BQ-018):

Correctly refused weather inquiry as unrelated to collections.

▣ PII Protection (BQ-023):

Declined bulk export of customer names and phone numbers.

FAIRNESS

Fairness Checks (BQ-029/031)

▣ Zero Discrimination:

Refused ranking by citizenship status or filtering by gender/marital.

▣ Regulatory Adherence:

Enforces non-negotiable Canadian Human Rights and Fair Lending rules.

FAIR LENDING

Protected Attributes Isolation

▣ Zero Demographic Leakage:

Gender, marital status, citizenship excluded from C360 and models.

▣ Fair Lending Compliance:

Affordability calculated strictly from verified financial facts.

INTEGRITY

Deterministic Calculations

▣ Zero LLM Math:

LLM is NEVER the calculator of record for balances, DSR, or surplus.

▣ DuckDB Math Engine:

All formulas deterministic, verifiable, and 100% reproducible.

AUDIT TRAIL

Enterprise Audit Trail

▣ Complete Logging:

Every ingestion, entity resolution link, and agent override logged.

▣ Policy Versioning:

All recommendations tied to immutable policy version (MAPLE-NBA-POL-1.0).

REGULATION

Regulatory Alignment

▣ Canadian Standards

Aligned with provincial collection rules, stay of proceedings, and DNCL.

▣ CASL Compliance:

Enforces servicing vs. marketing consent rules per POL-COLL-001.

SLIDE 10: BUSINESS IMPACT & FUTURE VISION

Quantifiable ROI and Strategic Roadmap for Maple Bank

EFFICIENCY

Frontline Efficiency

▣ Call Handling Time:

-40% reduction in average call duration via instant context delivery.

▣ Note Logging Time:

-75% reduction via automated interaction memory capture.

RECOVERY

Recovery Optimization

▣ +28% PTP Fulfillment

Sustainable, affordable payment plans are honoured far more reliably.

▣ -32% Roll-Rate to 90+:

Proactive early interventions prevent accounts deteriorating to default.

RETENTION

Borrower Retention

▣ Brand Preservation:

Respectful, empathetic engagement preserves valuable banking relationships.

▣ Cure to Advisory:

Enables seamless transition from collections to financial wellness.

VISION

Strategic Paradigm Shift

▣ From Dunning to Health:

Data Silos -> Case -> Aggressive Dunning TRANSFORMS INTO:

▣ Governed Closed Loop:

Golden ID -> C360 -> Memory -> Health -> Fair NBA -> Human Action ▣